

Name: TAYLOR'S VERSION

Student ID: _____

ECON 0150 | MiniExam 4 | Spring 2026

This MiniExam will take 20 minutes with a quick break to follow. MiniExams are designed to both test your knowledge and challenge you to apply familiar concepts in new environments. Treat it as if you're trying to show me that you understand the material. Answer clearly, completely, and concisely.

Academic Conduct Code The following academic conduct code is designed to protect the integrity of your work. Print your name/initials beside the three academic honesty agreements. I pledge to my fellow students, the university, and the instructor, that:

_____ I will complete this MiniExam solely using my own work.

_____ I will not use any digital resources unless explicitly allowed by the instructor.

_____ I will not communicate directly or indirectly with others during the MiniExam.

Q1. A researcher wants to understand whether mental health outcomes are related to wealth inequality. Using data from 60 US cities, they measure each city's Gini coefficient (gini, normalized to a between 0-100 where higher = more inequality) and average depression rate (depression_rate, cases per 1,000 residents).

$$\text{depression} = \beta_0 + \beta_1 \times \text{gini} + \epsilon$$

a) Write down (above) a statistical model to test this question.

b) Sketch (to the right ->) a visualization of this model.

c) Interpret an estimated coefficient of $\beta_1 = 0.48$ in words:

An increase of 1 gini is associated with 0.48 higher depression rate.

d) What is the default null hypothesis for β_1 ? 0



Q2. A school district tests whether a free tutoring program affects math scores recorded as scores. We code tutoring as 1 for students in the program and 0 for students not in the program. In a sample of 90 students, those in the tutoring program scored an average of 78 and those not in the program scored an average of 72.

$$\text{math} = \beta_0 + \beta_1 \times \text{tutor} + \epsilon$$

a) Write down (above) a statistical model to test this question.

b) Based on the information given, what would β_0 equal? 72

c) Based on the information given, what would β_1 equal? 6

d) If we instead coded **no_tutoring** as 1 for students NOT in the program and 0 for students in the program, what would β_0 equal? 78

Q3. A public health researcher studies whether air quality (PM2.5 concentration) predicts hospital admission rates per 100,000 people using data from $n = 120$ counties with the following output:

	coef	std err	t	P> t	[0.025	0.975]
Intercept	35.200	4.800	7.333	0.000	25.700	44.700
pm25	2.150	0.620	3.468	0.001	0.923	3.377

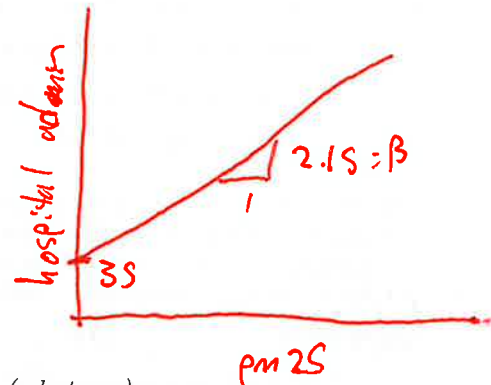
a) Sketch (to the right ->) how you would visualize the fitted model. Label the intercept and slope using the fitted values.

b) Interpret the Intercept coefficient (35.20) in context:

Every 1 unit increase in pm25 is associated with 2.15 more hospital admissions.

c) Interpret the pm25 coefficient (2.15) in context:

The rate of hospital admissions with no pm25 is roughly 35.2.



d) What does the p-value of 0.001 for the pm25 coefficient mean? (select one)

☒ If air quality has no effect on admissions, 0.1% of samples would have a coefficient this far from zero

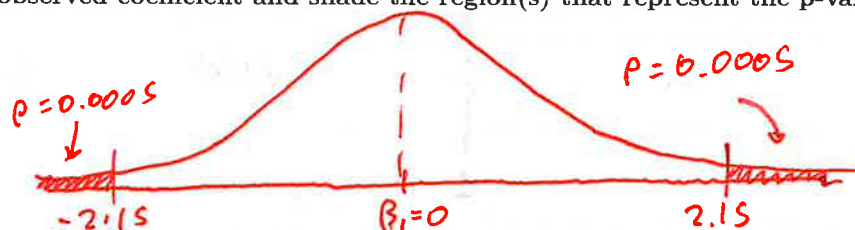
☐ Out of 1000 samples where air quality truly matters, only 1 would show a coefficient this large

☐ Only 0.1% of the hospital admissions difference is due to random chance

☐ There's a 99.9% probability that the true coefficient is at least 2.15

☐ The coefficient of 2.15 has a 0.1% margin of error

Q4. Draw the sampling distribution under the null hypothesis ($\beta_1 = 0$) for the pm25 coefficient from Q3. The observed coefficient is 2.15 with a standard error of 0.620 and a p-value of 0.001. Mark our observed coefficient and shade the region(s) that represent the p-value.



Q5. A researcher fits a regression model and examines the residual plot (predicted values on x-axis, residuals on y-axis). The plot shows residuals tightly clustered near zero for small predicted values but spreading out widely for large predicted values (a fan shape).

a) Which model assumption does this pattern suggest is violated?

Heteroskedasticity

b) Name one approach the researcher could use to address this issue

Robust standard errors.



MINIEXAM 4 RUBRIC

①

Q1. 20pts

5pts a) Excellent: correct 5pts

Satisfactory: flipped outcome and predictor: $gini = \beta_0 + \beta_1 \text{depression} + \epsilon$ 4pts

Progressing: wrote general form: $y = \beta_0 + \beta_1 x + \epsilon$ 2pts

Incomplete: something else 1pt

5pts b) ~~nothing~~

Excellent: scatter, regression line, gini on x, depression on y, 5pts
axes labeled

Satisfactory: one small error 4pts

Progressing: multiple small omissions or errors 2pts

Incomplete ~~nothing~~: ~~nothing~~ very little ~~to nothing~~ 1pt

5pts c) Excellent: 1) one-unit change in x 5pts

2) direction and magnitude of change

3) uses variable names and units

Satisfactory: one minor issue 4pts

Progressing: one major issue/omission or multiple minor issues 2pts

Incomplete ~~nothing~~: multiple major issues ~~or blank~~ 1pt

5pts d) Excellent: $\beta_1 = 0$ 5pts

Satisfactory: in words, like "no relationship" 4pts

Progressing/Incomplete: $\beta_1 \neq 0$ or something else 1pt

Blank: 0pt

Q2 20ptsSpts a) Same as Q1aSpts b) Spec: Student knows β_0 is the mean of the 0 group.Excell: 72 is correct 5ptsProgressing: 78 or 6 or 74 2ptsIncomplete: anything else 1ptSpts c) Spec: Student knows the difference between group means.Excell: $6 = 78 - 72$ 5ptsSatisfactory: -6, wrong sign 4ptsProgressing/Incomplete: 78 or 72 or 150 1ptSpts d) Spec: Student demonstrates that β_0 depends on which group is coded as 0.Excell: 78 5ptsProgressing/Incomplete: 72 or -6 or 6 2pts

likely memorized

Q3 20ptsSpts a) Spec: student can read a regression output, identify a numerical predictor model, draw a scatter and line, and label with fitted.Excell: 1) Scatter 5pts

2) Upward line

3) β_0 and β_1 labelsSatisfactory: one omission/error 4ptsProgressing: two omissions/errors 2ptsIncomplete: many issues 1pt

(3)

Sptsb) Spec: Student explains what β_0 means: the predicted y at $x=0$.Excellent: 1) When $x=0$ (ie concentration is zero)Spts2) Predicted y

3) Uses variable names or units

Incomplete: missing / incorrect item
one4ptsProgressing: ~~Incomplete~~ multiple issues2ptsSptsc) Spec: same structure as Q1c. Student explains what β_1 means: the average change in y per one-unit change in x .Spts

d) Correct: Spts

Incorrect: 0pts

Q4 15pts

Spec: Student can draw the null distribution, locate their observed statistic, and identify the p-values as tail areas.

Excellent: 1) Bell curve: symmetric, unimodal15pts

2) Centered at zero

3) Labeled x -axis as β_1 or 'slope'

4) Sample slope of 2.15 marked/labeled

5) Equally extreme -2.15 value marked

6) Shaded tails for p-values

Satisfactory: ~~two minor issues or one omission/issue~~
~~one major issue~~13ptsProgressing: two omissions/issues9ptsIncomplete: more than two omissions/issues5ptsBlank:0pts

Q5. 15pts

(4)

8pts a) Spec: Student correctly identifies the fan-shaped pattern as a violation of homoskedasticity.

Excellent: mentions homoskedasticity or heterosked. 2pts

Satisfactory: mentions something like "variance is not constant" 7pts

Progressing/Incomplete: anything else 2pts

7pts b) Spec: student mentions a valid remediation.

Excellent: robust standard errors OR log the y 7pts

Progressing: log the data or remove outliers 4pts

Incomplete: anything else 2pts